

Battery monitor IC with Coulomb counter/gas gauge

Features

- Battery voltage monitoring
- Internal temperature sensor
- Coulomb counter with 12/14-bit AD converter, +/- 80 mV input voltage range
- Internal or external 32768 Hz time base
- I2C interface for gas gauge monitoring and device control
- 32-RAM bytes
- 8-byte unique device ID
- One general-purpose I/O

Applications

- Cellular phones, PDA, MP3 players, cordless phones
- Digital cameras, USB appliances, Bluetooth devices

Description

The STC3100 monitors the critical parameters of a single-cell Li-Ion battery (voltage, temperature and current) and includes hardware functions to implement a gas gauge for battery charge monitoring, based on a programmable 12- to 14-bit A/D converter. With a typical 30 milliOhms external sense resistor, the battery current can be up to 2.5 A and the accumulator system provides a capacity up to +/-7000 mAh with a resolution of 0.2 mAh.

The device is programmable through the I2C interface.

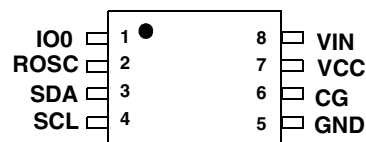


MiniSO-8
(Plastic micropackage)



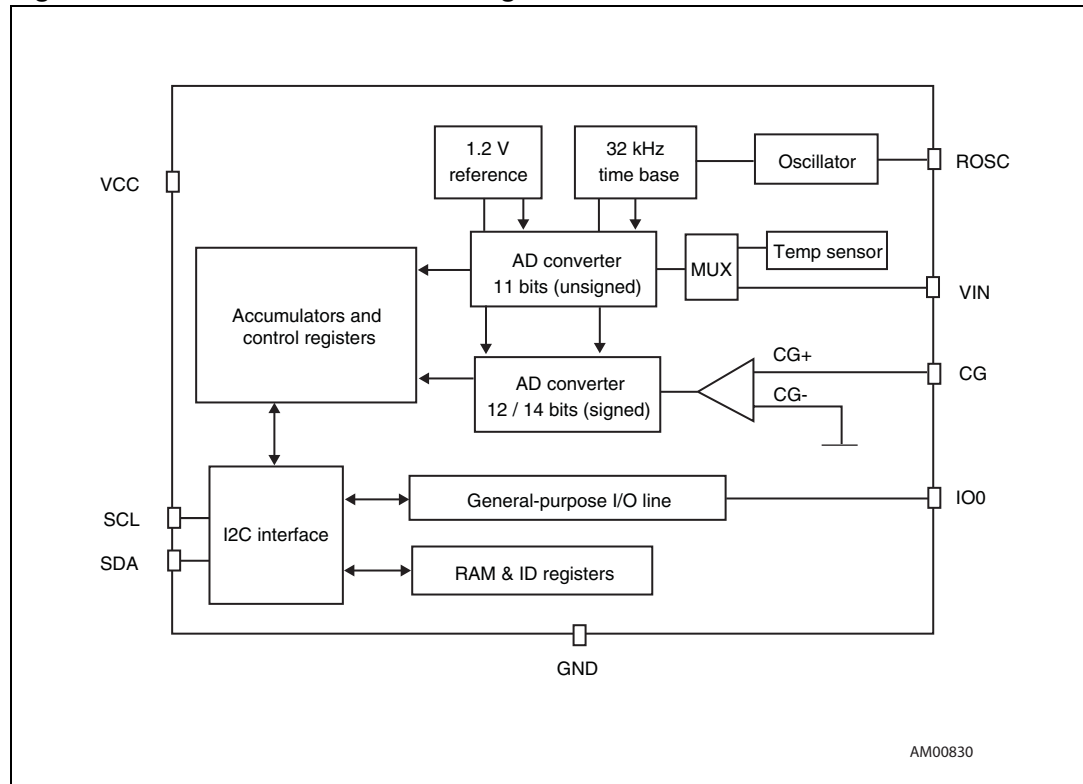
DFN8 3x3
(Plastic micropackage)

Pin connections
(top view)



1 Block diagram

Figure 1. STC3100 internal block diagram



2 Pin assignment

Table 1. STC3100 pin description

Pin #	Pin name	Type	Function
1	IO0	I/OD	General-purpose I/O
2	ROSC	I_AD	Oscillator bias resistor or external 32 kHz clock for gas gauge
3	SDA	I/OD	I2C serial data
4	SCL	I_D	I2C serial clock
5	GND	Ground	Analog and digital ground
6	CG	I_A	Gas gauge current sense input
7	VCC	Supply	Power supply
8	VIN	I_A	Battery voltage sense input

Note:

I: input

O: output

OD: open drain

A: analog

D: digital

3 Absolute maximum ratings and operating conditions

Table 2. Absolute maximum ratings

Symbol	Parameter	Value	Unit
V _{max}	Maximum voltage on any pin	7	V
V _{io}	Voltage on I/O pins	-0.3 to 7	V
T _{stg}	Storage temperature	-55 to 150	°C
T _j	Maximum junction temperature	150	°C
ESD	Electrostatic discharge (HBM human body model)	2	kV

Table 3. Operating conditions

Symbol	Parameter	Value	Unit
V _{cc}	Operating supply voltage on V _{CC}	2.7 to 5.5	V
V _{in}	Input voltage on V _{in}	0 to V _{cc}	V
V _{min}	Minimum voltage on V _{CC} for RAM content retention	2.0	V
T _{oper}	Operating free air temperature range	-40 to 85	°C

4 Electrical characteristics

Table 4. Electrical characteristics (2.7 V < V_{CC} < 4.5 V, -20° C to 70° C)

Symbol	Parameter	Conditions	Min	Typ	Max	Units
Supply						
I _{CC}	Operating current consumption	Average value over 4s			100	uA
I _{stby}	Current consumption in standby	standby mode, inputs=0V			2	uA
I _{pdn}	Current consumption in power down	V _{CC} < UVLO _{th} , inputs=0V			1	uA
UVLO _{th}	Undervoltage threshold	(V _{CC} decreasing)	2.5	2.6	2.7	V
UVLO _{hyst}	Undervoltage threshold hysteresis			100		mV
POR	Power-on reset threshold	(V _{CC} decreasing)		2.0		V
Gas gauge A/D converter						
V _{in_gg}	Input voltage range		-80		+80	mV
I _{in}	Input current for CG pin				500	nA
ADC _{res}	AD converter granularity	12 bits 13 bits 14 bits		47.08 23.54 11.77		uV uV uV
ADC _{offset}	AD converter offset	CG = 0V 12 bits 13 bits 14 bits	-2 -2 -3		2 2 3	LSB LSB LSB
ADC _{time}	AD conversion time (32768Hz clock)	12 bits 13 bits 14 bits		125 250 500		ms ms ms
ADC _{acc}	AD converter gain accuracy at full scale	25° C over temperature range			0.5 1	% %
F _{osc}	Internal time base frequency	R _{osc} = 200 kΩ 0.1%		32768		Hz
Osc _{acc}	Internal time base accuracy	25° C, V _{CC} = 3.6 V over temperature and voltage ranges			2 2.5	% %
F _{osc_ext}	External time base frequency range		30		70	kHz
Cur _{res}	Current register LSB value			11.77		uV
Chg _{res}	Charge register LSB value (32,768 Hz clock)	=Cur _{res} *2 ¹² *0.5/3600		6.70		uV.h
Global_ CG _{acc}	Gas gauge accuracy (not including the external sense resistor tolerance)	Using internal time base Using external time base			3.5 1	% %

Table 4. Electrical characteristics (2.7 V < V_{CC} < 4.5 V, -20° C to 70° C) (continued)

Symbol	Parameter	Conditions	Min	Typ	Max	Units
Battery voltage and temperature a/d converter						
Vin_adc	Input voltage range	V _{CC} = 5 V	0		5	V
LSB	LSB value	Voltage measurement Temperature measurement		2.44 0.125		mV °C
ADC_time	AD conversion time (32,768 Hz clock)			250		ms
Volt_acc	Battery voltage measurement accuracy	2.7 V < V _{in} < 4.5 V, V _{CC} = V _{in}	-0.5		+0.5	%
Temp_acc	Internal temperature sensor accuracy		-3		+3	°C
Digital I/O pins (SCL, SDA, IO0)						
Vih	Input logic high		1.2			V
Vil	Input logic low				0.35	V
Vol	Output logic low (SDA, IO0)	I _{OL} = 4 mA			0.4	V

5 Typical performance curves

Figure 2. Standby current vs. temperature

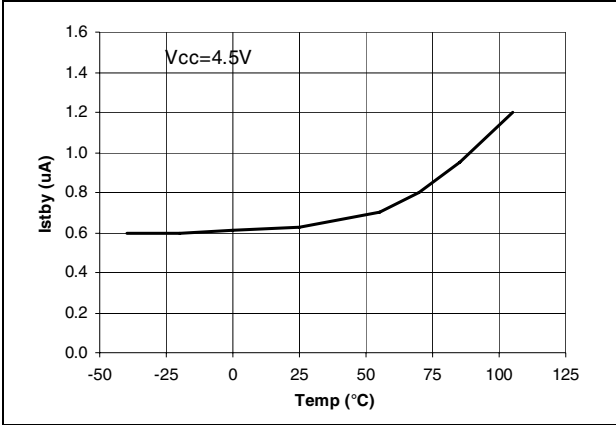


Figure 3. Current measurement accuracy vs. temperature

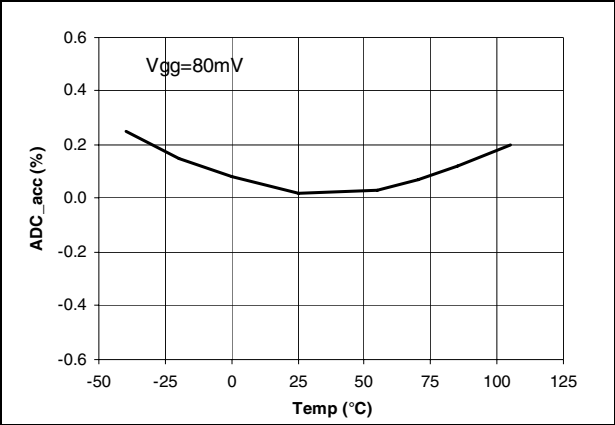


Figure 4. Oscillator frequency accuracy vs. temperature

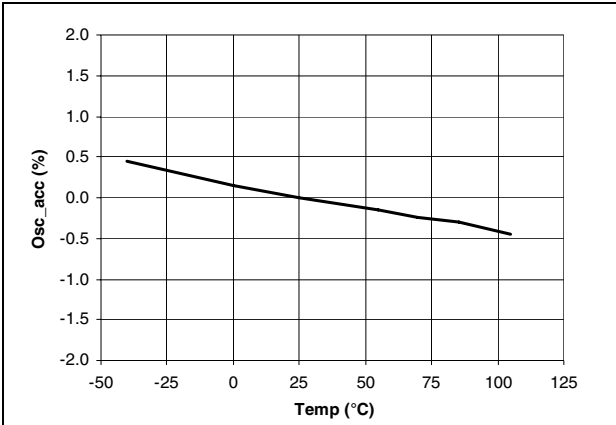
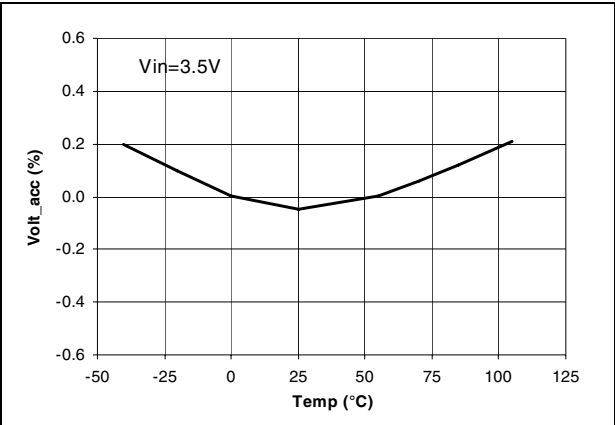


Figure 5. Voltage measurement accuracy vs. temperature



6 Application information

Figure 6. Example of an application schematic using the STC3100

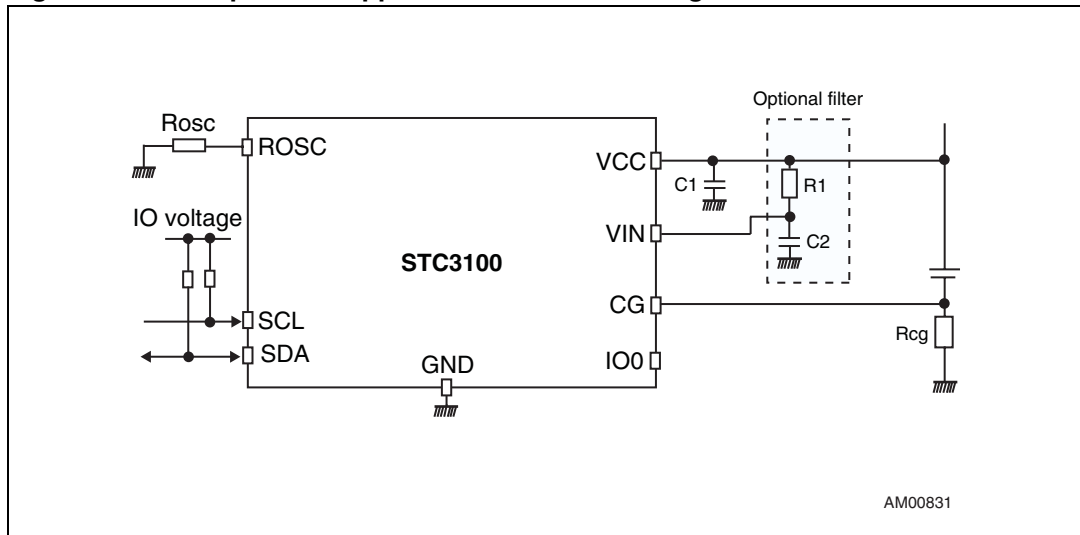


Table 5. External components list

Name	Value	Tolerance	Comments
R _{cg}	10 to 50 mΩ	1%	Gas gauge sense resistor
R _{osc}	200 kΩ	0.1%	Internal oscillator bias resistor
C1	1 μF		Supply decoupling capacitor
C2	220 nF		Battery voltage input filter (optional)
R1	1 kΩ		Battery voltage input filter (optional)

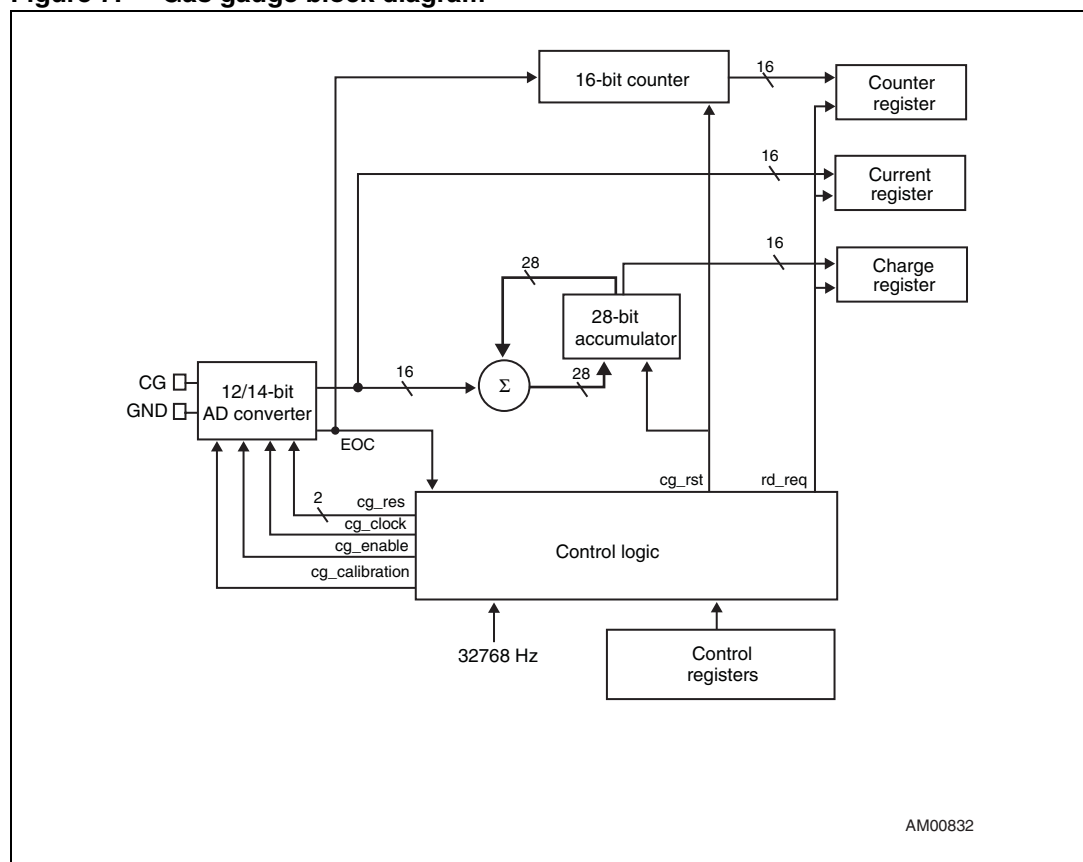
7 Functional description

7.1 Gas gauge

The gas gauge is used to monitor the available battery capacity. The voltage drop across the external sense resistor is integrated during a conversion period and input to a 12- to 14-bit AD converter. The output conversion is accumulated into a 28-bit accumulator. The system controller can control the gas gauge and read the data (upper 16 bits of the accumulator) through the I2C control registers.

The AD converter output is in two's complement format. When a conversion cycle is completed, the result is added to the charge accumulator and the number of conversions is incremented in a 16-bit counter.

Figure 7. Gas gauge block diagram



The controller can read the value of the most recent conversion in two's complement format by reading the REG_CURRENT registers. These registers are updated at the end of each conversion.

The differential inputs are scaled to the full range of the AD converter, introducing a small offset error. A high value written to the CG_CAL bit of the control register connects the inputs of the AD converter together, allowing the controller to measure the digital offset error. Using this measurement, one can calibrate the gas gauge and reduce errors due to the internal offset error.

The conversion cycle for n bit resolution is 2^n clock cycles. Using the 32,768 Hz internal clock, the conversion cycle time is 125 to 500 ms for a 12- to 14-bit resolution. The LSB value is set by the internal gain and internal reference and is 11.77 μ V at maximum resolutions. When using an external 30 milliOhms sense resistor, the 28-bit accumulator results in a capacity of approximately ± 7300 mA.h. The upper 16 bits of the accumulator can be read from the I2C interface, giving a resolution of 0.2 mA.h.

When the battery voltage falls below the under voltage lockout threshold at 2.7 V, the gas gauge system is stopped and the STC3100 stays in standby mode with minimum quiescent current. All registers are maintained down to 2.0 V. Below 2.0 V, the registers are reset to their default power-on value.

The gas gauge system needs an accurate 32,768 Hz timebase to compute the level of charge flowing from/to the battery. The STC3100 can operate from an internal oscillator, or use an external RTC signal for highest accuracy.

7.2 Battery voltage and temperature monitoring

The battery voltage and chip temperature (close to the battery temperature) are measured by means of an A/D converter and a multiplexer. This function takes place concurrently to the gas gauge function with a dedicated A/D converter, which means that it does not affect the performance of the gas gauge. To reduce the power consumption, a conversion takes place only every two seconds, alternatively for battery voltage and temperature (so each value is refreshed every four seconds).

The conversion cycle takes $2^{13} = 8192$ clock cycles. Using the 32,768 Hz internal clock, the conversion cycle time is 250 ms. The resolution is 2.44 mV for the battery voltage and 0.125° C for the temperature.

7.3 General-purpose input/output

A general-purpose I/O line is available. The output is an open drain, and an external pull-up resistor may be needed in the application. Writing the IO0DATA bit to 0 forces the IO0 output low; writing the IO0DATA bit to 1 leaves the IO0 output in a high impedance state. Reading the IO0DATA bit gives the state of the IO0 pin.

In standby (CG_RUN=0), reset (PORDET set to 1) and power-down ($V_{cc} < UVLO_{th}$) states, the IO0 output is open and the input is read as zero whatever is the actual state of the IO0 pin.

7.4 RAM registers

The STC3100 provides 32 RAM registers to store any information regarding battery status, charge cycles, battery aging, proprietary informations, etc...

The register content is maintained during standby and low voltage states, down to the power-on reset level of approximately 2.0 V. Below this level, the content is not preserved. This usually means that the Li-Ion cell was very deeply discharged and has been damaged.

7.5 Unique device ID

The STC3100 provides a means to identify the battery pack or the subsystem. Each device has its own unique 8-byte ID made of an 8-bit part ID (value = 10h for the STC3100), a 48-bit random unique ID and an 8-bit CRC.

The CRC-8 is calculated according to bytes REG_ID0 to REG_ID6 using the " $x^8 + x^2 + x + 1$ " polynomial with a zero initial value.

Since the device ID is downloaded from the ROM at power-up and is subsequently kept in read-only RAM locations together with the general-purpose RAM registers, the device ID can also be used as an indicator of the RAM integrity.

